

TECHNICAL SPECIFICATIONS

SECTION 26 05 00

COMMON ELECTRICAL REQUIREMENTS

Project: Meridian Data Centre Campus, Phase 1
Navi Mumbai, India
Revision: 2024 (December 12, 2024)

PART 1 - GENERAL

1.1 Summary

This section specifies the common requirements, quality assurance, submittal procedures, and certification standards for all Division 26 electrical work. Other sections in Division 26 point to this section as the central reference hub.

1.2 References

- IS 732 - Code of practice for electrical wiring installations
- IS 3043 - Code of practice for earthing
- IS 1646 - Code of practice for fire safety of buildings (general)
- IEC 62040-3 - Uninterruptible power systems (UPS) - Method of specifying performance

1.3 Submittal Requirements

A. Contractor shall submit the following for Engineer's approval prior to equipment procurement:

1. Product Data: Technical datasheets detailing all electrical and physical parameters, performance curves, and compliance.
2. Shop Drawings: Dimensional layouts showing clearance requirements, cabling paths, and weights.
3. Compliance Matrix: Clause-by-clause confirmation of conformance. Any deviation must be explicitly noted.
4. Factory Test Certificates and Type Test Certifications.

1.4 Quality Assurance

A. Equipment shall be manufactured by a firm specializing in the equipment class with a minimum of twenty (25) years of continuous commercial manufacturing history.

B. As specified in Section 26 05 00 Part 1.4.3.B, all medium voltage switchgear and high-power electrical components must be supplied with certified test certificates from an internationally recognized independent laboratory (ASTA, KEMA, or equivalent) confirming compliance with IEC standards.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

A. Subject to compliance with technical requirements, approved manufacturers are limited to the following:

2.2 Performance and Design Requirements

general_testing: All electrical equipment must be type-tested. Independent ASTA/KEMA certificates are mandated for medium voltage gear as specified in Part 1.4.3.B.

submittal_procedures: Submittal files must be uploaded in searchable PDF format and include a compliance matrix.

PART 3 - EXECUTION

3.1 Installation and Examination

- A. Verify that structural concrete floor slab is fully cured. Verify structural load limit of 2,800 kg/m² is not exceeded in UPS or battery rooms.
- B. Maintain a minimum clear work clearance of 1000 mm in front of all enclosures, and 1200 mm above, in accordance with local Indian Electricity Rules and IEC standards.

3.2 Field Quality Control and Site Commissioning

- A. Startup services shall be performed by factory-authorized technicians.
- B. Commissioning tests must be scheduled and carried out in accordance with the project's overall Cx Test Register. Every testing step must map back to a specific specification clause.

grounding: Verify all earthing connections are tight. Measure loop resistance using calibrated earth tester.

quality_control: Contractor must perform certified field tests prior to final inspection.